Machine Learning Study Notes

Sample material for testing the Adaptive Learning Chatbot.

Chapter 1: Introduction to Machine Learning

Machine learning (ML) is a branch of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed with rules. A machine learning model is trained on examples, and it then uses what it learned to make predictions on new, unseen data.

There are three main types of machine learning: supervised learning, unsupervised learning, and reinforcement learning. Each type differs in the kind of data it uses and the feedback it receives during training.

Supervised learning uses labeled data. Every training example contains an input and the correct output. For instance, a set of emails labeled 'spam' or 'not spam' can be used to train a spam filter. Common supervised tasks are classification, where the output is a category, and regression, where the output is a number such as a house price.

Unsupervised learning uses unlabeled data. The algorithm tries to discover hidden structure on its own. Clustering is a common example: a shop can group customers with similar buying habits without knowing the groups in advance. Dimensionality reduction, such as PCA, is another unsupervised technique that compresses data into fewer features.

Reinforcement learning trains an agent that interacts with an environment. The agent receives rewards for good actions and penalties for bad ones, and it learns a policy that maximizes the total reward over time. Game-playing programs and robot controllers often use this approach.

Chapter 2: Training and Evaluating Models

A dataset is usually divided into three parts: the training set, the validation set, and the test set. The training set is used to fit the model. The validation set is used to tune hyperparameters. The test set is used only once at the end to measure how well the model generalizes.

Overfitting happens when a model learns the training data too closely, including its noise, and therefore performs poorly on new data. Underfitting happens when a model is too simple to capture the real pattern. Techniques to reduce overfitting include collecting more data, using regularization, applying dropout in neural networks, and stopping training early.

Accuracy is the fraction of predictions that are correct. When classes are imbalanced, accuracy can be misleading, so precision, recall, and the F1-score are used. Precision is the fraction of predicted positives that are truly positive. Recall is the fraction of actual positives that the model found. The F1-score is the harmonic mean of precision and recall.

Cross-validation gives a more reliable estimate of performance. In k-fold cross-validation the data is split into k parts. The model is trained k times, each time using a different part as the validation set and the remaining parts for training. The k scores are then averaged.

Chapter 3: Gradient Descent and Neural Networks

Gradient descent is an optimization algorithm used to minimize a loss function. The loss function measures how far the model's predictions are from the true values. At each step, the algorithm computes the gradient of the loss with respect to the model parameters and moves the parameters a small step in the opposite direction.

The size of that step is controlled by the learning rate. If the learning rate is too large, the algorithm may overshoot the minimum and fail to converge. If it is too small, training becomes very slow. Variants such as stochastic gradient descent and Adam improve speed and stability.

An artificial neural network is made of layers of connected units called neurons. Each neuron computes a weighted sum of its inputs, adds a bias, and passes the result through an activation function such as ReLU or sigmoid. Activation functions introduce non-linearity, which allows the network to learn complex patterns.

Backpropagation is the algorithm used to compute gradients in a neural network. It applies the chain rule of calculus to propagate the error from the output layer back to the earlier layers, so that every weight can be updated. A network with many hidden layers is called a deep neural network, and training such networks is known as deep learning.